

ULIP Integration Document

Version No.: 4.0

Date: 07/04/2026

Project Name: ULIP

This document contains proprietary information of NLDSL. Unauthorized access, copying and replication are prohibited. This document must not be copied in whole or part by any means, without the written authorization of NLDSL, Noida, India.

Revision History

Version	Date	Author	Reviewer	Significant Changes
1.0	21/03/2024	Hasan Mhad Khan	Nayab Ali	Baseline Version
2.0	01/10/2024	Ayush Arya	Nayab Ali	Updated
3.0	07/03/2025	Ayush Arya	Nayab Ali	Request body changes
4.0	07/04/2026	Gaurav Tyagi	Pranab Manna	Updated

Table of Contents

Introduction	4
1.1 Purpose	4
The Overall Description	4
1.2 Data Integration between EVYATRA System and ULIP application	4
1.3 EVYATRA/01	4
1.3.1 Technical Approach	6
1.4 Data Transmission & Authentication Mechanism	9

Introduction

1.1 Purpose

The purpose of this document is to define and provide details for accessing API for EV charging stations across India, via using input parameters such as state and geolocation such as latitude and longitude.

This document details the following points:

1. **Data Integration between Bureau of Energy Efficiency (BEE) and ULIP web service:** Identification of operational points where data integration between **Bureau of Energy Efficiency (BEE)** and ULIP web service will be needed.
2. **Technical Approach** for Performing Data Integration

The Overall Description

Users will need to share their state ID or geolocation to access the list of public electric vehicle charging stations through the rest API. A comprehensive list of state ID can be found by clicking on the provided link.

Here are some points to keep in mind when accessing EV charging station information:

- The API will only share data that is based on the user's provided information.
- ULIP will provide an API that allows users to obtain data from the BEE API.
- The EVYATRA/01 API will retrieve data from the BEE database using the stateId, or geolocation (latitude and longitude) provided by the user.

1.2 Data Integration between EVYATRA System and ULIP application

The ULIP system will receive information from the user and provide data in return.

1.3 EVYATRA/01

This API is designed to retrieve information about electric vehicle (EV) charging stations. It can be used by providing either the state ID or the geolocation (latitude and longitude) of the desired location.

Please note that when making a request, **all of the JSON keys are mandatory in the payload**. If you choose to use the stateId to access EV stations, the latitude and longitude fields should be left empty.

```
{
  "stateId": "1",
  "lat": " ",
  "long": " "
}
```

Use the state code below from the table to inquire about the (EV) charging stations outlet in the respective state.

State Name	State ID
Andaman & Nicobar Islands	1
Andhra Pradesh	2
Arunachal Pradesh	3
Assam	4
Bihar	5
Chandigarh	6
Chhattisgarh	7
Delhi	10
Goa	11
Gujarat	12
Haryana	13
Himachal Pradesh	14
Jammu & Kashmir	15
Jharkhand	16
Karnataka	17
Kerala	18
Lakshadweep	19
Madhya Pradesh	20
Maharashtra	21
Manipur	22
Meghalaya	23
Mizoram	24
Nagaland	25
Odisha	26
Puducherry	27
Punjab	28
Rajasthan	29
Sikkim	30
Tamil Nadu	31

Tripura	32
Telangana	33
Uttar Pradesh	34
Uttarakhand	35
West Bengal	36
Ladakh	37
UT OF D&NH AND D&D	38
Lakshadweep	44

When sending geolocation data, if the user provides geocoordinate latitude and longitude, the state ID will be empty.

```
{
  "stateId": "",
  "lat": "11.604053",
  "long": "92.694109"
}
```

	Field Name	Field Description	Format	Required/Not – Required	Length
1	stateId	State ID	[0-9]{0,6}	Required	0-6
2	lat	latitude	^[0-9]{2}\.{0,1}[0-9]{2,10}[[0-9]{0,2}	Required	0-13
3	long	longitude	^[0-9]{2}\.{0,1}[0-9]{2,10}[[0-9]{0,2}	Required	0-13

1.3.1 Technical Approach

1. All the data will be shared through REST web services.
2. The user will share the input for which information is required.
3. EVYATRA API will share details of EV charging station information that exists in the BEE system in JSON format.
4. The ULIP system will provide data in JSON Object format in the response body.

Example:

Request: The request must be of the following format, where **state ID, latitude and longitude** are passed within the URL post request body in JSON format.

For production –

<https://www.ulip.dpiit.gov.in/ulip/v1.0.0/EVYATRA/01>

```
{
  "stateId": "",
  "lat": "18.595556",
  "long": "73.717167"
}
```

Curl URL

```
curl -L -X POST 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/EVYATRA/01' -H 'Accept: application/json' -H 'Content-Type: application/json' -H 'Authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWliOiJ1bGlwliwiaWF0IjoxNzEyMjA1MzIxLCJhcHBzljoiZGF0YXB1c2gifQ.O9k3Lg_L1e55hatY-54V7tK-ctMsDy5gNjuLQeAja3RbXctpz_r_TTvzmZgj_v-z7i9tifH_LpDKZvYoOW12mw' --data-raw '{
  "stateId": "",
  "lat": "18.595556",
  "long": "73.717167"
}'
```

For staging –

<https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/EVYATRA/01>

```
{
  "stateId": "",
  "lat": "18.595556",
  "long": "73.717167"
}
```

Curl URL

```
curl --location 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/EVYATRA/01' \
--header 'Authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWliOiJ1bGlwMlslmlhdCI6MTcyNzY5MDAwMCwiYXBwcyI6ImRhdGFwdXNoIn0.2BrY8dCCS_BAH5Y6faXPYAPJEV52bGuZZRtKXD04AOmAUtooBy9TC-wmWVvxyAWsqLR9ifBhYD9xfqGMDPH9_A' \
--header 'Content-Type: application/json' \
--header 'Accept: application/json' \
--data '{
  "stateId": "",
  "lat": "18.503194",
  "long": "73.969694"
}'
```

For obtaining an authentication token, refer to [Section 1.6](#)

Responses

To acknowledge the above request, ULIP will send the below response -

a) In case of an invalid format

1. Invalid stateld

```
{
  "response": null,
  "error": "true",
  "code": "400",
  "message": "Data format failed OR wrong value entered at: stateld. Format should follow [0-9]{0,6}$"
}
```

2. Invalid lat and long format

```
{
  "response": null,
  "error": "true",
  "code": "400",
  "message": "Data format failed OR wrong value entered at: lat. Format should follow ^[0-9]{2}\.\{0,1\}[0-9]{2,10}|[0-9]{0,2}$"
}
```

b) In case EV charging station data does not exist in the EVYATRA system

```
{
  "response": [
    {
      "response": [],
      "responseStatus": "SUCCESS",
      "message": null
    }
  ],
  "error": "false",
  "code": "200",
  "message": "Success"
}
```

c) In case EV charging stations exist

```
{
  "response": [
    {
      "response": [
        {
          "id": 31290,
          "cpo_id": 1144,
```

```
{
  "state_id": 21,
  "city_id": 6602,
  "discom_id": null,
  "station_name": null,
  "address": "22/11 Kv Ibp Switching S/Stn",
  "pincode": null,
  "lat": 18.595556,
  "lng": 73.717167,
  "createdBy": null,
  "createdDateGnrl": 0,
  "isActive": 0,
  "isDeleted": 0,
  "lastUpdatedDateTimezoneld": 0,
  "lastUpdatedDateGnrl": 0,
  "updatedBy": null
},
{
  "responseStatus": "SUCCESS",
  "message": null
}
],
{
  "error": "false",
  "code": "200",
  "message": "Success"
}
```

Status Code 200: EV charging station data response

Status Code 400: Bad request (Invalid JSON syntax, invalid JSON key)

Status Code 401 or 403: Unauthenticated or Unauthorized

Status Code 500: Some internal server errors occurred

Status Code 502: The server is not responding.

1.4 Data Transmission & Authentication Mechanism

All data exchanges will be securely done over HTTPS. ULIP's system will request data from EVYATRA through a URL with the following format:

For staging -

<https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/EVYATRA/01>

Body:

```
{
  "stateId": "",
  "lat": "18.503194",
  "long": "73.969694"
}
```

To access the data exchange URL, you would need to use a security mechanism that is implemented by the ULIP system. The process is as follows:

- First time access: you would require a username and password. The username and password would be shared with you beforehand. Once you enter the valid username and password, the ULIP system will return a basic authorization token to you. You can use this token to communicate with the ULIP system until the token gets expired.

In brief, HTTP Basic Authentication works as follows:

When you hit the ULIP secured API for the first time, you need to provide a username and password to get an access token.

```
curl --location --request POST
```

```
'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/user/login' \
--header 'Accept: application/json' \
--header 'Content-Type: application/json' \
--data-raw '{
"username": "xxxx",
"password": "xxxx@123"
}'
```

Authorization:

Bearer

eyJhbGciOiJIUzUxMiJ9.eyJzdWl0OiJ0ZXN0MSIsImh0dCI6MTYyNTIzMDQyNywiYXBwcyI6ImRhZGFwdXN0In0.rhumwWJ8BWeeHe9CYmIV1UKyJk-kLHA_ovX5Zi2OwxEv-td-KpxnfbKDI_8FZsOogggqx3Al-Du1y2qMoUD7uzA'

After successfully authenticating, the user will be logged into the ULIP system and obtain an access token. This token will be used by the user to communicate with the ULIP system. The token has an expiry time that represents the user session, which is typically set to 24 hours. If there is no request from the user within this time frame, the session (token) will expire, and the user will need to re-authenticate.

With the obtained token, the user can request EV charging station details from the ULIP system as follows:

```
curl -L -X POST 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/EVYATRA/01' -H 'Accept: application/json' -H 'Content-Type: application/json' -H 'Authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ1bGwliwiaWF0IjoxNzEyMjA1MzIxLCJhcHBzljoiZGF0YXBX1c2gifQ.O9k3Lg_L1e55hatY-54V7tK-ctMsDy5gNjuLQeAja3RbXctpz_r_TTvvmZgj_v-z7i9tifH_LpDKZvYoOW12mw' -data '{"stateId": "", "lat": "18.503194", "long": "73.969694"}'
```

For production –

```
curl --location 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/user/login' \  
--header 'Accept: application/json' \  
--header 'Content-Type: application/json' \  
--data-raw '{  
  "username": "xxxx",  
  "password": "xxxx@123"  
}'
```

Authorization:**Bearer**

```
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ1bGluIiwiaWF0IjoxNzI3MTYxMDgwLCJhcHBzIjoizGF0YXB1c2gifQ.XhVZm7NiCD05-v9091WJQhYzdk1a0Shs_Ley0T4R2MX63swRyxHbHxh6bVXtzqWvWiSLwwNGRQIcqGhr49hpGg
```

Upon successful authentication, the user would be logged into the ULIP system and get an access token. Though this token user can communicate with ULIP. This token has expiry time which represent user session. The session time out for the same to be fixed hour, generally it is for 30 minutes. If there is no request coming from user for 30 minutes, then session (token) will be expired and user need to re authenticate.

Using The above token, user can request ULIP system for details as follow -

<https://www.ulip.dpiit.gov.in/ulip/v1.0.0/EVYATRA/01>

```
curl --location 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/EVYATRA/01' \  
--header 'Authorization: Bearer  
eyJhbGciOiJIUzUxMiJ9.eyJhcHBzIjoizGF0YXB1c2gifQ.XhVZm7NiCD05-v9091WJQhYzdk1a0Shs_Ley0T4R2MX63swRyxHbHxh6bVXtzqWvWiSLwwNGRQIcqGhr49hpGg'  
--header 'Accept: application/json' \  
--header 'Content-Type: application/json' \  
--data '{  
  "stateId": "",  
  "lat": "18.503194",  
  "long": "73.969694"  
}'
```